

Answers: Introduction to t -testing

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Summary

Answers to a selection of questions for the study guide on introduction to t -testing.

These are the answers to [Questions: Introduction to \$t\$ -testing](#).

Please attempt the questions before reading these answers.

Q1

- 1.1. Sample size less than 30.
- 1.2. A hypothesis test where the test statistic follows the t distribution.
- 1.3. A t -test with equal variances.
- 1.4. A t -test with unequal variances.

Q2

- 2.1. $\bar{x}$ is the sample mean. μ_0 is the known value. s is the sample standard deviation. n is the sample size.
- 2.2. μ_1 is sample mean 1. μ_2 is sample mean 2. s_1 is sample standard deviation 1. s_2 is sample standard deviation 2. n_1 is sample size 1. n_2 is sample size 2.
- 2.3.
 - a) $s_1 = s_2$
 - b) $s_1 \neq s_2$

Q3

- 3. 1. $H_0 : \bar{x} = \mu_0 H_1 : \bar{x} > \mu_0$
- 3.2. $H_0 : \mu_1 = \mu_2 H_1 : \mu_1 > \mu_2$
- 3.3. $H_0 : \mu_1 = \mu_2 H_1 : \mu_1 \neq \mu_2$
 $(10 - 7)/(s/\sqrt{19}) = 26.1534$

Student's t -test.

One sample t -test are always Student t -tests.

$$(12 - 13)/((2/\sqrt{15}) + (2/\sqrt{16})) = -0.9839$$

Student's t -test.

$$s_1 = s_2$$

$$(45 - 51)/((3/\sqrt{19}) + (3.5/\sqrt{18})) = -3.9651$$

Welch's t -test.

$$s_1 \neq s_2$$

Version history and licensing

v1.0: initial version created 03/26 by Millie Harris (as part of a University of St Andrews VIP project)

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